

Software Engineering (CS24353) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

EXECUTIVE MASTER STUDY GUIDE & CASE STUDY BANK

Software Engineering (CS24309)

Birla Institute of Technology, Mesra | B.Tech CSE 5th Semester (NEP 2024-25 Scheme)

Complete Course Structure & Engineering Lifecycle Matrix

Module	Core Syllabus Scope	Key Methodologies & Formulations
Module I	Process Models & Agile	Waterfall, Evolutionary Prototyping, Boehm Spiral, Agile Scrum, Sprint Burndown, Extreme Programming, CMMI
Module II	Requirements & Modeling	Functional/Non-Functional Requirements, IEEE 830 SRS Standard, Use Case Modeling, Class Diagrams, Statecharts
Module III	Design & Architecture	High Cohesion & Low Coupling Metrics, Architectural Styles (Layered, MVC, Microservices), GoF Design Patterns
Module IV	Software Testing & QA	Black-Box (Equivalence Partitioning, BVA), White-Box (Basis Path, McCabe Cyclomatic Complexity $V(G) = E - N + 2P$), Mutation
Module V	Estimation & Management	COCOMO II Cost Estimation Equations, Function Point (FP) Analysis, CPM/PERT Critical Path, Risk Management

Exam Preparation & High-Yield Strategy

This publication-grade master book consolidates all 5 modules with formal software estimation formulas, step-by-step worked Cyclomatic Complexity & Function Point numericals, and comprehensive model answers to BIT Mesra end-semester examination questions.

Module I: Process Models & Agile Methodologies — Topics Covered

1. Software Crisis & Layered Technology
2. Prescriptive Lifecycle Models (Waterfall, Spiral, RAD)
3. Agile Manifesto & Scrum Framework
4. Extreme Programming (XP & TDD)
5. Project Scheduling & Risk Management (RMMM)

1. Software Process Lifecycle Models Comparison

Model	Core Characteristics	Ideal Project Profile
Waterfall Model	Strict linear sequential phases. High documentation discipline. No customer feedback until late testing phase.	Well-understood, stable requirements (e.g., compilers, aerospace flight control).
Spiral Model (Boehm)	Risk-driven evolutionary meta-model combining prototyping with systematic waterfall milestones across 4 quadrants.	Large, expensive, high-risk, mission-critical systems.
Scrum (Agile)	Iterative, time-boxed Sprints (2–4 weeks), daily stand-ups, self-organizing teams, shippable product increments.	Rapidly evolving requirements, web/mobile applications.

Module II: Requirements Engineering & IEEE 830 SRS — Topics Covered

1. Functional vs. Non-Functional Requirements
2. Requirements Elicitation (FAST, JAD, Interviews)
3. Data Flow Diagrams (DFD Level 0, 1, 2)
4. IEEE 830–1998 Standard SRS Structure
5. Requirements Traceability & Validation

1. IEEE Standard 830–1998 Structure of SRS

- **1. Introduction:** Purpose, Scope, Definitions, References, Overview.
- **2. Overall Description:** Product perspective, Functions, User classes, Operating environment, Design constraints, Assumptions.
- **3. Specific Requirements:** Functional requirements, Performance criteria, Security/Safety constraints, External interface specifications (UI, Hardware, Software, Communications).

Module III: Software Architecture, Cohesion & UML — Topics Covered

1. Architectural Concepts & SOLID Principles
2. Modularity: Cohesion (7 Levels) & Coupling (6 Levels)
3. Architectural Styles (Layered, Microservices, MVC)
4. UML Structural: Use Case & Class Diagrams
5. UML Behavioral: Sequence, Activity & Statecharts

1. Cohesion and Coupling Quality Spectrum

Desirable Design Goal: High Cohesion within modules and Low (Loose) Coupling between modules.

Cohesion Spectrum (Low → High / Desirable)	Coupling Spectrum (High / Undesirable → Low / Desirable)
1. Coincidental (Worst)	1. Content Coupling (Worst – direct memory modification)
2. Logical	2. Common Coupling (Shared global variables)
3. Temporal	3. External Coupling
4. Procedural	4. Control Coupling
5. Communicational	5. Stamp Coupling (Passing composite data structure)
6. Sequential	6. Data Coupling (Best – passing scalar parameters)
7. Functional Cohesion (Best – performs single focused task)	

Module IV: Verification, Validation & Software Testing — Topics Covered

1. Verification vs. Validation (Boehm's Criterion)
2. Black-Box Testing: Equivalence Partitioning & BVA
3. White-Box Testing: McCabe's Cyclomatic Complexity
4. Levels of Testing: Unit, Integration (Stubs/Drivers), System
5. Software Reliability: MTTF, MTBF & Availability

1. McCabe's Cyclomatic Complexity $V(G)$ Formulations

Given a Control Flow Graph (CFG) G :

1. $V(G) = E - N + 2$ (where E = edges, N = nodes)
2. $V(G) = P + 1$ (where P = predicate / decision nodes with multiple outgoing branches)
3. $V(G)$ = Number of enclosed planar regions R

BIT Mesra Exam Question (10 Marks)

Problem: Draw the CFG and compute Cyclomatic Complexity for finding max of 3 numbers: `if (a > b) { if (a > c) max = a; else max = c; } else { if (b > c) max = b; else max = c; }`

Solution: The code contains 3 predicate/decision nodes ($P = 3$). Thus, $V(G) = P + 1 = 3 + 1 = 4$. There are 4 linearly independent execution basis paths requiring at least 4 test cases for 100% branch coverage.

Module V: Project Estimation, COCOMO & CMMI — Topics Covered

1. Function Point Analysis (Albrecht UFP & VAF)
2. COCOMO I (Organic, Semidetached, Embedded Modes)
3. SEI CMMI 5–Level Maturity Framework
4. Software Configuration Management (SCM Baselines)
5. Software Maintenance: 4 Types & Lehman's Laws

1. Basic COCOMO Estimation Formulas (Barry Boehm)

$$\text{Effort (Person-Months)} = a_b \times (\text{KLOC})^{b_b}$$

$$\text{Development Time (Months)} = c_b \times (\text{Effort})^{d_b}$$

Project Mode	Team & Project Description	a_b	b_b	c_b	d_b
Organic	Small experienced team, well-understood domain (< 50 KLOC).	2.4	1.05	2.5	0.38
Semidetached	Medium team, mixed experience, medium size (50 – 300 KLOC).	3.0	1.12	2.5	0.35
Embedded	Tight hardware/software constraints, mission-critical.	3.6	1.20	2.5	0.32